

Energy Efficiency Through Education and Low-Cost Measures

There are many benefits to combining energy efficiency with energy education. The strategy is cost-effective, and it can reach many people with a low investment of time and money. It helps participants understand how energy is used in their homes and empowers them to generate their own energy savings. We have found, through our work as economic consultants for energy efficiency programs, that effective energy education includes client-specific messages; an action focus; a highly interactive atmosphere with hands-on learning opportunities; the translation of energy impacts to dollar savings; written commitments from clients; and follow up with participants. The states that have implemented education programs have all been highly satisfied with the outcomes. (For more on energy education, see "Gizmo Giveaways Reduce Energy Use," *HE* July/Aug 2003, p. 8.)

Our company, Quantec, recently evaluated energy efficiency programs in Iowa, South Carolina, and Indiana that apply successful education strategies. Integrated data collection was an element of each of the programs. Participants reported on baseline consumption characteristics, the installation of energy efficiency measures, and the adoption of savings actions. The savings estimates were drawn from these data, and the results have been impressive.

Table 1. Iowa Average Participant Savings

Measure Installation	Electric (kWh)	Gas (therms)	Water (gallons)
CFL—1	80		
CFL—2	75		
Showerhead	101	20.7	3,731
Faucet aerator—kitchen	30	6.1	1,577
Faucet aerator—bathroom	46	9.4	2,431
Filter tone alarm	8	15.8	
Education Impact			
Adjust hot water heater	17	4.2	
Adjust heating	58	31.5	
Adjust air conditioning	17		
Adjust refrigerator or freezer	8		
Reduce hot water use	22	3.5	1,040
Total Savings	462	91.3	8,779

Table 2. South Carolina Average Participant Savings

Measure Installation	Electric (kWh)	Gas (therms)	Water (gallons)
CFL—1	28		
CFL—2	34		
Showerhead	330	5.4	3,417
Faucet aerator—kitchen	89	1.1	1,183
Faucet aerator—bathroom	195	2.4	2,592
Filter tone alarm	48	4.4	
Education Impact			
Adjust hot water heater	67	1.0	
Adjust heating	161	13.4	
Adjust air conditioning	26		
Adjust refrigerator or freezer	8		
Reduce hot water use	90	0.8	1,040
Total Savings	1,076	28.5	8,231

Iowa Utilities Association

Iowa's investor-owned utilities (Aquila, Alliant Energy-IPL, and MidAmerican Energy) have chosen to make a significant investment in energy education, coupled with low-cost energy efficiency measures for low-income households in the state. Their annual investment of approximately \$60,000 includes an energy efficiency

kit delivered to each household, a train-the-trainers workshop, an energy education workshop curriculum, and an agency administrative payment of \$10 per participant. This investment has yielded a combined \$136,000 in annual savings for the 1,000 households that participated, or an average participant savings of \$126 (see Table 1). The bulk of the dollar savings for participants in Iowa was realized in lower natural gas bills. The saturation of natural gas is quite high in Iowa; between 80% and 90% of the program participants use natural gas for water and/or space heating. The rest use electricity, oil, and/or wood.

The energy efficiency kit delivered to participants in Iowa contained a variety of low-cost energy efficiency tools: two compact fluorescent lamps, a low-flow showerhead, faucet aerators for both the kitchen and bathroom sinks, and a furnace filter tone alarm. The kit also included several tools to help customers implement energy efficiency practices in their households, such as thermometers to measure freezer/refrigerator settings and temperature in the home and tools to measure the flow rate for faucets and showerheads. These items work to reinforce participants' knowledge of the sources of their energy costs and, consequently, how to reduce those costs.

South Carolina Department of Energy

State and tribal Low Income Home Energy Assistance Program (LIHEAP) programs are block grant programs; LIHEAP grantees have a certain amount of discretion in how they spend their federal LIHEAP dollars. All LIHEAP grantees sign certain Assurances, that is, provisions regarding the operation of the LIHEAP program. Assurances deal with eligibility, benefit level, administrative costs, and other program factors. Assurance 16, in particular, allows states and tribes the option of using up to 5% of their assistance funds for low-income households to reduce their home energy needs and promote self-sufficiency. Under Assurance 16, states are able to spend money on energy education programs coupled with low-cost energy efficiency measures.

The South Carolina Governor's Office, Office of Economic Opportunity (OEO) in South Carolina instituted its energy efficiency program using its Assurance 16 dollars from the Department of Health and Human Services. The cost to the OEO of South Carolina for serving 1,200 households was \$60,000. The South Carolina program has generated excellent installation rates among participating households. The average savings per household is \$123 per year in utility costs, for a combined program total of \$147,000 (see Table 2). The saturation of natural gas usage in South Carolina is 20% for water heating and 34% for space heating. This is a lower saturation than in Iowa, yet the total per-participant dollar savings generated is quite similar.

The measures included in the kit participants received varied little from those in the Iowa kits. The South Carolina program does have a unique approach to securing postinstallation customer surveys—it sends commu-

nity action agency staff into the field to follow up with participants. Due to the great success of the program, the OEO of South Carolina is encouraging local agencies to utilize Assurance 16 dollars for the energy efficiency and education program for the 2005/2006 program year.

State of Indiana

The State of Indiana created a Conservation Action Kit, the contents of which qualified as weatherization measures, making it eligible for federal

ana is considering expanding the program to reach non-LIHEAP-eligible families in future years by leveraging other funding sources, while continuing the incentives the program offers to encourage participation.

The energy education programs in Iowa, South Carolina, and Indiana used different funding approaches, but with common educational elements, and achieved similarly impressive results. Across each program

about two-thirds of the annual savings comes from installation of low-cost measures and the other third comes from energy efficiency practices adopted by participants. These savings from the measures will continue over the measure life, while the change in behaviors can continue indefinitely with proper reinforcement and encouragement.



—M. Sami Khawaja
and Jill E. Steiner

Table 3. Indiana Average Participant Savings

Measure Installation	Electric (kWh)	Gas (therms)	Water (gallons)
CFL—1	95		
CFL—2	66		
Showerhead	224	7.4	4,309
Faucet aerator—kitchen	59	1.9	1,030
Faucet aerator—bathroom	29	1.0	669
Filter tone alarm	28	1.9	
Education Impact			
Adjust hot water heater	31	1.9	
Adjust heating	108	27.2	
Adjust air conditioning	55		
Adjust refrigerator or freezer	1		
Reduce hot water use	696	41.3	6,008
Total Savings	1,393	82.6	12,016

DOE weatherization dollars. The Indiana program has continued since 2003, serving over 25,000 households and saving, on the average, \$102 per participant (see Table 3).

Indiana's investment in the program has been astounding. In addition to the Conservation Action Kit, it decided to give agencies a \$25-per-participant incentive to deliver the program. The participants also received a \$25 credit to their utility bill when completed survey materials were returned. A telephone survey conducted after the program determined that participants were able to easily recall the energy efficiency practices that they had learned about in the course of the energy education training.

The breadth of households reached by the program was so large that Indi-

M. Sami Khawaja is the president and Jill E. Steiner is a project director at Quantec, LLC, in Portland, Oregon, a consulting firm specializing in economic issues in energy, energy efficiency, demand side management, education, and health services.

For more information:

Quantec, LLC
720 SW Washington St., Ste. 400
Portland, Oregon 97205
Tel: (503)228-2992
Web site: www.quantecllc.com